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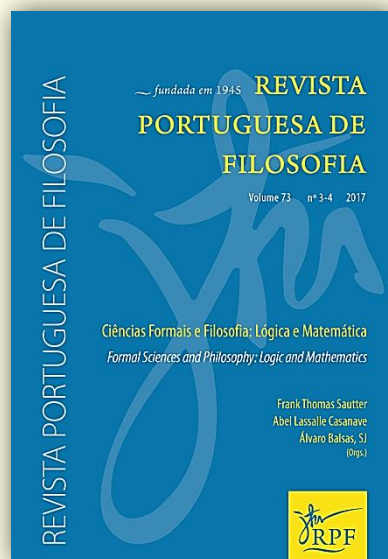
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Normalization, Soundness and Completeness for the Propositional Fragment of Prawitz' Ecumenical System*

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Abstract

In 2015 Dag Prawitz proposed an Ecumenical system where classical and intuitionistic logic could coexist in peace. The classical logician and the intuitionistic logician would share the universal quantifier, conjunction, negation and the constant for the absurd, but they would each have their own existential quantifier, disjunction and implication, with different meanings. Prawitz' main idea is that these different meanings are given by a semantical framework that can be accepted by both parties. The aim of the present paper is [1] to prove the normalization theorem for the propositional fragment NE_p of Prawitz' ecumenical system, and [2] to show that NE_p is sound and complete with respect to a Kripke-style semantics for the language of NE_p .

Keywords: classical logic, ecumenical logic, intuitionistic logic.

1. Introduction

It is a fact that for more than two millennia a single logic prevailed, Aristotelian logic, the logic that obeyed some fundamental principles, such as the principle of non-contradiction (the supreme principle of all principles) and the principle of the third-excluded middle. Even with the fregean revolution of the late nineteenth century, this situation didn't change, at least till the beginning of the twentieth century: a single logic, classical logic. However, it is also a fact that this state of affairs has changed drastically in the last hundred years: several logics have presented themselves as extensions or rivals of classical logic. For example, the addition of

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several operators of modal nature to the so-called classical logic produced modal logics of various types (alethic modal logics, epistemic logics, deontic logics, temporal logics, computational logics). Such logics can be considered as extensions of classical logic in the sense that they only widen the scope of logical-conceptual analysis. On the other hand, the revision of the unrestricted validity of certain fundamental principles produced a set of logics that presented themselves as real alternatives to classical logic, as rivals of classical logic. Thus, for example, intuitionist logic questioned the unrestricted validity of principle of the excluded middle, and several paraconsistent logics questioned the unrestricted validity of the principle of non-contradiction. But how can one question and propose a revision of fundamental principles such as the principle of the excluded middle and the principle of non-contradiction? How can such deep disagreements arise in the realm of logic? An interesting and robust argument against the possibility of revisionism in logic is based on a remark made by Quine in 1970¹. The argument can be roughly presented as follows.

If the revisionist/deviant logician does not accept the general validity of a classical principle of reasoning P, then the revisionist logician assigns new meanings to the concepts (or at least to some of the concepts) used in the formulation of the principle P. If the revisionist logician assigns new meanings to the concepts used in the formulation of the principle, then the revisionist logician and the classical logician are not talking about the same thing, about the same principle. So the classical logician and the intuitionist logician cannot disagree!

In a recent paper, Dag Prawitz seems to agree with Quine, when he says:

When the classical and intuitionistic codifications attach different meanings to a constant, we need to use different symbols, and I shall use a subscript c for the classical meaning and i for the intuitionistic.²

But, according to Prawitz, to assert that the classical and the intuitionistic codifications attach different meanings to some constant and to recognize the need to use different symbols corresponding to these

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1. Willard van Orman Quine, *The Philosophy of Logic*. New Jersey: Prentice-hall, 1970.
 2. Dag Prawitz, "Classical versus intuitionistic logic". In *Why is this a proof*, edited by Edward Hermann Haeusler, Wagner Sanz and Bruno Lopes, (UK: College Books, 2015), 28

different meanings is not a kind of trivial acceptance of Quine's position, for, as Prawitz puts it,

This does not imply that the classical meanings of these constants cannot be explained in the same general way as the intuitionistic meanings of the logical constants have been explained.³

In order to explore and develop this idea, Prawitz introduces: (1) a language with some constants that are common to classical and intuitionistic logic ($\perp$, $\neg$, $\wedge$, and $\forall$), constants that are classical ($\vee_c$, $\rightarrow_c$ and $\exists_c$), and constants that are intuitionistic ($\vee_i$, $\rightarrow_i$, and $\exists_i$); (2) an ecumenical natural deduction system⁴ in which, as we said, "the classical and intuitionistic constants can then have a peaceful coexistence in a language that contains both". The aim of the present paper is [1] to prove the normalization theorem for the propositional fragment NE_p of Prawitz' ecumenical system, and [2] to show that NE_p is sound and complete with respect to a Kripke-style semantics for the language of NE_p .

2. The Natural Deduction Ecumenical system NE_p

The propositional language of the ecumenical logic E_p is defined as follow:

Alphabet

1. Propositional letters: $p, q, r, \dots$
2. Logical constants: $\perp, \wedge, \neg, \vee_i, \vee_c, \rightarrow_i$ and $\rightarrow_c$.
3. Auxiliary signs: $(,)$.

The grammar for the language of E_p is inductively defined in the usual way. Besides "pure" classical and "pure" intuitionistic formulas, we can also have "hybrid" formulas in the language, as for example, $((\neg p \vee_c q) \rightarrow_i r)$.

The Natural Deduction system NE_p defined by Prawitz has the following rules of inference:

1. The rules for $\wedge$, $\neg$ and for the intuitionistic operators are the usual Gentzen

3. Prawitz, *op. cit.*, 26.

4. Prawitz, *op. cit.*, 28-29.

– Prawitz introduction and elimination rules for these operators.

2. The intuitionistic absurd rule:

$$\frac{\perp}{A}$$

Where A is a propositional variable.

3. The rules for classical implication and classical disjunction are defined as follows:

$$\begin{array}{c} [A] \qquad \qquad \qquad [\neg B] \\ \Pi_1 \\ \frac{\perp}{A \rightarrow_c B} \quad \rightarrow_c - \text{Int} \\ \frac{A \rightarrow_c B \quad A \quad \neg B}{\perp} \quad \rightarrow_c - \text{Elim} \end{array}$$

$$\begin{array}{c} [\neg A] \qquad \qquad \qquad [\neg B] \\ \Pi_1 \\ \frac{\perp}{A \vee_c B} \quad \vee_c - \text{Int} \\ \frac{A \vee_c B \quad \neg A \quad \neg B}{\perp} \quad \vee_c - \text{Elim} \end{array}$$

Notations:

1. Π/A denotes a derivation Π with conclusion A.
2. A/Π denotes a derivation Π with assumptions of the form of A.
3. $\Pi_1/[A]/\Pi_2$ denotes the result of replacing the assumptions A in Π_2 by the derivation Π_1/A .
4. $r[\Pi]$ denotes the last rule of inference applied in Π .

Brief remarks on classical implication and negation:

- (1) Classical implication: Contrary to what we could expect from any reasonable concept of conditional judgments (hypothetical judgments), the operator $\rightarrow_c$ does not *unconditionally* satisfy modus ponens. This is due to the fact that the introduction rule for $\rightarrow_c$ is weaker than the

introduction for $\rightarrow_i$, since the classical logician is allowed to assert $(A \rightarrow_c B)$ in cases where the intuitionistic logician is not allowed to assert $(A \rightarrow_i B)$. It is interesting to observe that the general validity of modus ponens for $\rightarrow_c$ does not depend solely on the meaning of $\rightarrow_c$, but depends also on a concept of negation that is not determined by the introduction rule for negation. The classical implication $\rightarrow_c$ clearly satisfies a weak form of modus ponens: $\{A, (A \rightarrow_c B)\} \vdash \neg\neg B$.

- (2) Negation: It would be natural to ask why the system has just one negation, given that it has two implications and the $\perp$. The point is that $(A \rightarrow_i \perp)$ is equivalent to $(A \rightarrow_c \perp)$, both classically and intuitionistically. In a more general sense, one could also say that there is just one way to assert a negated proposition: assume the proposition and show that this assumption leads to a contradiction.

Some interesting theorems

1. $\vdash NE_p (A \rightarrow_i B) \rightarrow_i (A \rightarrow_c B)$
2. $\vdash NE_p (A \wedge B) \leftrightarrow_i \neg(\neg A \vee_c \neg B)$
3. $\vdash NE_p (A \wedge B) \leftrightarrow_i \neg(A \rightarrow_c \neg B)$
4. $\vdash NE_p \neg(\neg A \wedge \neg B) \leftrightarrow_i (A \vee_c B)$
5. $\vdash NE_p \neg(A \wedge \neg B) \leftrightarrow_i (A \rightarrow_c B)$

Prawitz observes that a formula containing only classical operators has a “classical behavior”, i.e., that it satisfies all the principles of classical logic, as for example, modus ponens and the classical *reductio*.⁵ In fact we can keep part of the classical behavior with “hybrid” formulas, i.e., with formulas that contains classical and intuitionistic operators.

Definition 1. A formula B is called *externally classical* if and only if its main operator is classical (we can indicate that B is externally classical with the notation B^c)

We can now have some other interesting theorems

1. $\vdash NE_p (A \rightarrow_c B^c) \rightarrow_i (A \rightarrow_i B^c)$
2. $\{[A, (A \rightarrow_c B^c)] \vdash B^c\}$.

As we can see, preservation of modus ponens for classical implication depends only on the consequent being externally classical.

5. See, Prawitz, *op. cit.*, 29.

3. For every A and B, $(A \rightarrow_c B)$ and $(A \vee_c B)$ satisfy the classical reductio.

For example, assume that

$$\begin{array}{c} \neg(A \rightarrow_c B) \\ \Pi \\ \perp \end{array}$$

We can produce the effect of the classical reductio with the following derivation:

$$\begin{array}{c} \frac{[(A \rightarrow_c B)]^1 \quad [A]^2 \quad [\neg B]^3}{\perp} \\ 1 \frac{\perp}{\neg(A \rightarrow_c B)} \\ \Pi \\ 2,3 \frac{\perp}{(A \rightarrow_c B)} \end{array}$$

3. Normalization for the system NE_p

3.1 Detours and Reductions

The introduction and elimination rules of the system NE_p may produce *detours*, as in the case of classical and intuitionistic logic. We may introduce a formula by an application of an introduction rule and immediately after use this formula as the major premiss of an elimination rule. Some detours of this kind may be hidden by applications of $\forall_i$ elimination. In this subsection we officially introduce these detours and the operations whose aim is to eliminate them.

Definition 2. A formula A in a derivation Π is a maximum formula if and only if

1. A is the major premiss of an application of an α -introduction rule and at the same time the major premiss of an application of an α -elimination rule.
2. A is the conclusion of an application of $\forall_i$ -elimination and at the same time major premiss of an application of an elimination rule.

Definition 3. As usual, we have some operations, the so-called *reductions*,

whose aim is to locally eliminate maximum formulas. The reductions for $\wedge$, $\neg$ and for the intuitionistic operators are the usual Prawitz-reductions (including the permutative reductions). The reductions for the classical operators are defined below:

$$\frac{\begin{array}{c} [A] \\ \Pi_1 \\ \hline \frac{\perp}{A \rightarrow_c B} \end{array} \quad \begin{array}{c} [\neg B] \\ \Pi_2 \quad \Pi_3 \\ A \quad \neg B \end{array}}{\perp}$$

Reduces to:

$$\frac{\begin{array}{c} \Pi_2 \quad \Pi_3 \\ [A] \quad [\neg B] \\ \Pi_1 \\ \perp \end{array} \quad \begin{array}{c} [\neg A] \quad [\neg B] \\ \Pi_1 \quad \Pi_3 \\ \hline \frac{\perp}{A \vee_c B} \end{array}}{\perp}$$

Reduces to:

$$\frac{\begin{array}{c} \Pi_2 \quad \Pi_3 \\ [\neg A] \quad [\neg B] \\ \Pi_1 \\ \perp \end{array}}{\perp}$$

If a derivation Π is transformed into a derivation Π' by means of one or more reductions, we say that Π reduces to Π' . In the case of the new reductions we immediately see that through the elimination of a maximum formula, new maximum formulas of the same degree may be produced, and because of this, the usual normalization strategies do not work anymore. An easy way to solve this difficult is through the modification of the usual definition of the degree of a formula as the number of occurrences logical operators in the formula. It is clear that in the case of classical disjunction and classical implication there are some hidden nega-

tions, and that any definition of the complexity of a formula must take this point in consideration. The new measure of complexity of a formula A will be called the ecumenical degree of A , $ed(A)$, and is defined as follows:

1. $ed(\varphi) = 0$, for any propositional variable φ .
2. $ed(\neg A) = ed(A) + 1$
3. $ed(A \wedge B) = ed(A) + ed(B) + 1$, if $\wedge$ is $\wedge$ or an intuitionistic operator.
4. $ed(A \vee_c B) = ed(A) + ed(B) + 4$
5. $ed(A \rightarrow_c B) = ed(A) + ed(B) + 3$

Definition 4. The ecumenical degree of a derivation Π , $ed(\Pi)$, is defined as $\max\{ed[A]: A \text{ is maximum formula in } \Pi\}$.

Definition 5. A derivation Π is normal if and only if $ed[\Pi] = 0$.

Definition 6. A derivation Π is called critical if and only if

1. Π ends with an application of an elimination rule α .
2. The major premiss A of α is a maximum formula.
3. $ed[\Pi] = ed[A]$.
4. For every proper sub-derivation Π' of Π , $ed[\Pi'] < ed[A] = ed[\Pi]$.

3.2 The Normalization Theorem

The strategy we will use to prove the normalization theorem for NE_p is due to Pottinger.⁶ Before we present the proof, we need some preparatory lemmas.

Lemma 1. Let Π_1/A and A/Π_2 be two derivations in NE_p such that $ed(\Pi_1) = n_1$ and $ed(\Pi_2) = n_2$. Then, $ed[(\Pi_1/[A]/\Pi_2)] = \max\{ed[A], n_1, n_2\}$.

Proof. Induction on the length of Π_2 .

Lemma 2. If Π reduces to Π' , then $ed[\Pi'] \leq ed[\Pi]$.

Proof. Directly from the form of the reductions and the definition of the complexity measure “ed”.

6. See, Garrel Pottinger, “A New Way of Normalizing Intuitionist Propositional Logic”. *Studia Logica* 35, no. 4 (1976): 387-408.

Lemma 3. Let Π be a critical derivation of $\Gamma \vdash A$. Then, Π reduces to a derivation Π' of $\Delta \subseteq \Gamma \vdash A$ such that $d(\Pi') < d(\Pi)$.

Proof: By induction on the length of Π .

- The major premiss of $r[\Pi]$ is the result of an introduction rule. Then, the result follows directly from the form of the reductions. We shall examine just one case, the other cases being treated in a similar way. Let Π be the following derivation:

$$\frac{\begin{array}{c} [\neg A] \\ \Pi_1 \\ \frac{\perp}{A \vee_c B} \end{array} \quad \begin{array}{c} [\neg B] \\ \Pi_2 \quad \Pi_3 \\ \neg A \quad \neg B \end{array}}{\perp}$$

The derivation Π reduces to the following derivation Π' :

$$\frac{\begin{array}{c} \Pi_2 \quad \Pi_3 \\ [\neg A] \quad [\neg B] \\ \Pi_1 \\ \perp \end{array}}{\perp}$$

By lemma [1], $ed[\Pi'] = \max\{ed[\Pi_2], ed[\Pi_3], ed[\neg A], ed[\neg B]\} < ed[\Pi]$.

- The major premiss of $r[\Pi]$ is the result of an application of $\vee_i$ -elimination.

The derivation Π has the following form:

$$\frac{\begin{array}{c} \Pi_1 \quad \Pi_2 \quad \Pi_3 \\ A \vee_i B \quad C \quad C \quad \Pi_4 \quad \Pi_5 \\ \frac{C}{D} \quad D \quad D \end{array}}{D}$$

The derivation Π reduces to the following derivation Π^* :

$$\frac{\begin{array}{c} \Pi_1 \quad \Pi_2 \quad \Pi_4 \quad \Pi_5 \quad \Pi_3 \quad \Pi_4 \quad \Pi_5 \\ A \vee_i B \quad \frac{C \quad D \quad D}{D} \quad \frac{C \quad D \quad D}{D} \end{array}}{D}$$

- If $ed[\Pi^*] < ed[\Pi]$, then we take $\Pi^* = \Pi'$.

- If $\text{ed}[\Pi^*] = \text{ed}[\Pi]$, then at least one of the derivations of the minor premises of $r[\Pi^*]$ is critical (remember that, given that Π is critical, $\text{ed}[(A \vee_i B)] < \text{ed}[C]$). For the sake of the argument we will assume that both derivations of the minor premises are critical.

By the induction hypothesis, the derivation

$$\frac{\Pi_2 \quad \Pi_4 \quad \Pi_5}{C \quad D \quad D} \quad D$$

reduces to a derivation Π_2' of D such that $\text{ed}[\Pi_2'] < \text{ed}[\Pi]$. The same holds for the derivation of the other minor premiss of $r[\Pi^*]$. We then take Π' to be the following derivation:

$$\frac{\Pi'_1 \quad \Pi'_2 \quad \Pi'_3}{(A \vee_i B) \quad D \quad D} \quad D$$

Lemma 4. Let Π be a derivation of $\Gamma \vdash A$ such that $\text{ed}[\Pi] > 0$. Then, Π reduces to a derivation Π' of $\Delta \subseteq \Gamma \vdash A$ such that $\text{ed}[\Pi'] < \text{ed}[\Pi]$.

Proof. Directly from lemmas [2] and [3] using induction on the length of Π .

Theorem 1. Let Π be a derivation of $\Gamma \vdash A$. Then, Π reduces to a normal derivation Π' of $\Delta \subseteq \Gamma \vdash A$.

Proof. Directly from lemma [4] by induction on the degree of Π .

Remark 1: Normal proofs in NE_p do not satisfy the usual properties we expect normal proofs to satisfy. For example, it is easy to see that NE_p does not satisfy the sub-formula principle:

$$\frac{\frac{[\neg A] \quad [\neg \neg A]}{\perp}}{(A \vee_c \neg A)}$$

This derivation is normal, but the formula $\neg \neg A$ is not a sub-formula of the conclusion $(A \vee_c \neg A)$.

Remark 2: Although NE_p does not satisfy the sub-formula property, the system does satisfy another important property usually attached to normal proofs: if Π is a closed derivation of A (no open assumptions), the $r[\Pi]$ is an application of an introduction rule.

4. Ecumenical Kripke Semantics

Prawitz' main philosophical motivation for the definition of the ecumenical system was to provide a constructive rule-based semantics for classical operators, but given the ecumenical nature of NE_p , it would be natural to approach it not only from a (constructive) semantics based on rules, but also from the more traditional notion of semantic satisfiability. In this section we define a Kripke-style semantics for the logic E_p and we prove the soundness and completeness of the system NE_p with respect to this semantics.

Definition 7. An ecumenical Kripke model is a pair $\langle F, e \rangle$ where $F = \langle W, \leq \rangle$ is an intuitionistic Kripke frame and e is a intuitionistic valuation, i.e., e is a mapping associating with each propositional variable p and element of $U_\leq$, where $U_\leq$ is the set of all uppersets of $\langle W, \leq \rangle$, i.e., all sets $U \subseteq W$ such that if $\omega \in U$ and $\omega \leq \vee$, then $\vee \in U_\leq$.

Let $M = \langle F, e \rangle$ be an ecumenical Kripke model and $\omega \in W$. By induction in the construction of a formula φ we define the relation $(M, \omega) \models \varphi$ as follows:

- $(M, \omega) \models p$ iff $\omega \in e(p)$;
- $(M, \omega) \not\models \perp$
- $(M, \omega) \models \neg\psi$ iff for all $v \in W$ such that $\omega \leq v$: $(M, v) \not\models \psi$;
- $(M, \omega) \models \varphi \wedge \psi$ iff $(M, \omega) \models \varphi$ and $(M, \omega) \models \psi$;
- $(M, \omega) \models \varphi \vee_i \psi$ iff $(M, \omega) \models \varphi$ or $(M, \omega) \models \psi$;
- $(M, \omega) \models \varphi \vee_c \psi$ iff $(M, \omega) \models \neg(\neg\varphi \wedge \neg\psi)$;
- $(M, \omega) \models \varphi \rightarrow_i \psi$ iff for all $v \in W$ such that $\omega \leq v$: $(M, v) \models \varphi$ implies $(M, v) \models \psi$;
- $(M, \omega) \models \varphi \rightarrow_c \psi$ iff $(M, \omega) \models \neg(\varphi \wedge \neg\psi)$.

The consequence relation $\models_{E_p}$ for the logic E_p is defined as usual: $\Gamma \models_{E_p} A$ if and only if every model of Γ is also a model of A .

4.1 Soundness and Completeness

In this subsection we prove that the system NE_p is sound and complete with respect to the ecumenical Kripke-semantics.

4.1.1 Soundness

Theorem 2 (Soundness).

$$\Gamma \vdash \text{NE}_p A \rightarrow \Gamma \models E_p A$$

Proof. The proof of soundness is carried out by induction on the length of derivations. We shall only consider the case $\vee_c$ -introduction, the other cases being treated in a similar way.

(1) $r[\Pi]$ is $\vee_c$ -introduction. The derivation Π has the following form:

$$\frac{\begin{array}{c} [\neg A] \quad \Gamma \quad [\neg B] \\ \Pi_1 \\ \perp \\ (A \vee_c B) \end{array}}{\perp}$$

By the induction hypothesis, we know that $\{\Gamma, \neg A, \neg B\} \models E_p \perp$, and this means that there is no world ω such that $(M, \omega) \models E_p \Gamma \cup \{\neg A\} \cup \{\neg B\}$. Now, assume that $\Gamma \not\models E_p (A \vee_c B)$, which is equivalent to say that $\Gamma \not\models E_p \neg(\neg A \wedge \neg B)$. According to the definition of semantical consequence, $\Gamma \not\models E_p \neg(\neg A \wedge \neg B)$ if and only if there exists a world v such that $(M, v) \models E_p \Gamma$ and a world v' such that $v' > v$ and $(M, v') \models E_p (\neg A \wedge \neg B)$. But, this leads to a contradiction, because for this world v' , $(M, v') \models E_p \Gamma \cup \{\neg A\} \cup \{\neg B\}$.

4.1.2 Completeness

In order to prove the completeness theorem for the system NE_p , we shall define a translation Pr from the language of E_p into the language of intuitionistic propositional logic.

1. $\text{Pr}[\varphi] = \varphi$, if φ is a propositional variable.
2. $\text{Pr}[\perp] = \perp$
3. $\text{Pr}[\neg A] = \neg \text{Pr}[A]$
4. $\text{Pr}[(A \wedge B)] = (\text{Pr}[A] \wedge \text{Pr}[B])$.
5. $\text{Pr}[(A \vee_i B)] = (\text{Pr}[A] \vee_i \text{Pr}[B])$.
6. $\text{Pr}[(A \rightarrow_i B)] = (\text{Pr}[A] \rightarrow_i \text{Pr}[B])$.
7. $\text{Pr}[(A \vee_c B)] = \neg(\neg \text{Pr}[A] \wedge \neg \text{Pr}[B])$.

$$8. \quad \text{Pr}[(A \rightarrow_c B)] = \neg(\text{Pr}[A] \wedge \neg\text{Pr}[B]).$$

Lemma 5. $\Gamma \vdash \text{NE}_p A$ if and only if $\text{Pr}[\Gamma] \vdash \text{I}_p \text{Pr}[A]$, where $\vdash \text{I}_p$ is the usual derivability relation in the propositional fragment of Prawitz' natural deduction intuitionistic system I.

Proof. From left to right: By induction over the length of the derivation Π of $\Gamma \vdash \text{NE}_p A$. We shall only consider the case where the last rule applied in Π is $\rightarrow_c$ -introduction, the other cases being treated in a similar way. Let Π be the following derivation in NE_p :

$$\begin{array}{c} [A] \quad [\neg B] \quad \Gamma \\ \Pi_1 \\ \hline \perp \\ (A \rightarrow_c B) \end{array}$$

By the induction hypothesis, we have a derivation $\text{Pr}[\Pi_1]$ of $\text{Pr}[\perp]$ from $\text{Pr}[A]$, $\text{Pr}[\neg B]$ and $\text{Pr}[\Gamma]$. We can then take our derivation $\text{Pr}[\Pi]$ in the system I to be:

$$\begin{array}{c} \frac{[(\text{Pr}[A] \wedge \neg \text{Pr}[B])]}{\text{Pr}[A]} \quad \frac{\frac{[(\text{Pr}[A] \wedge \neg \text{Pr}[B])]}{\neg \text{Pr}[B]}}{\text{Pr}[\neg B]} \quad \text{Pr}[\Gamma] \\ \text{Pr}[\Pi_1] \\ \hline \frac{\text{Pr}[\perp]}{\neg(\text{Pr}[A] \wedge \neg \text{Pr}[B])} \\ \text{Pr}[(A \rightarrow_c B)] \end{array}$$

In order to prove the other direction, we shall prove an auxiliary lemma:

Lemma 6. $\vdash \text{NE}_p (A \leftrightarrow_i \text{Pr}[A])$

Proof. The proof is carried out by induction on $\text{ed}[A]$. We shall examine two cases, the other cases being treated similarly.

(1) A is $\neg B$.

$$\begin{array}{c} \frac{[B] \quad B \leftrightarrow_i \text{Pr}[B]}{\text{Pr}[B]} \quad \text{Pr}[\neg B] \\ \hline \neg \text{Pr}[B] \\ \hline \perp \\ \neg B \end{array}$$

$$\frac{\frac{[\text{Pr}[B]] \quad (B \leftrightarrow_i \text{Pr}[B])}{B} \quad \neg B}{\perp} \quad \neg \text{Pr}[B]$$

$$\text{Pr}[\neg B]$$

(2) A is $(B \rightarrow_c C)$.

$$\frac{\frac{[\neg C] \quad \neg C \leftrightarrow_i \text{Pr}[\neg C]}{\text{Pr}[\neg C]} \quad \frac{[B] \quad B \leftrightarrow_i \text{Pr}[B]}{\text{Pr}[B]} \quad \neg \text{Pr}[C] \quad \text{Pr}[(B \rightarrow_c C)]}{\frac{(\text{Pr}[B] \wedge \neg \text{Pr}[C]) \quad \neg (\text{Pr}[B] \wedge \neg \text{Pr}[C])}{\perp} \quad (B \rightarrow_c C)}$$

$$\frac{\frac{(\text{Pr}[B] \wedge \neg \text{Pr}[C])}{\text{Pr}[B]} \quad (B \leftrightarrow_i \text{Pr}[B]) \quad \frac{(\text{Pr}[B] \wedge \neg \text{Pr}[C])}{\neg \text{Pr}[C]} \quad \frac{\neg \text{Pr}[C] \quad (\neg C \leftrightarrow_i \text{Pr}[\neg C])}{\neg C} \quad (B \rightarrow_c C)}{\frac{\perp}{\neg (\text{Pr}[B] \wedge \neg \text{Pr}[C])} \quad \text{Pr}[(B \rightarrow_c C)]}$$

The direction right to left now follows from the lemma above and the fact that $\text{Pr}[\Gamma] \vdash \text{I}_p \text{Pr}[A]$ implies that $\text{Pr}[\Gamma] \vdash \text{NE}_p \text{Pr}[A]$,

Lemma 7. If $\Gamma \models E_p A$, then $\text{Pr}[\Gamma] \models \text{I}_p \text{Pr}[A]$, where $\models \text{I}_p$ is the usual consequence relation of propositional intuitionistic logic.

Proof.

1. Assume that $\Gamma \models E_p A$.
2. By Lemma [6], $\forall C \in \Gamma \vdash \text{NE}_p (C \leftrightarrow_i \text{Pr}[C])$.
3. By the soundness of NE_p , we have $\forall C \in \Gamma \models E_p (C \leftrightarrow_i \text{Pr}[C])$.
4. Again, by Lemma [6] and soundness, we have $\models E_p (A \leftrightarrow_i \text{Pr}[A])$.
5. From which it follows that $\text{Pr}[\Gamma] \models E_p \text{Pr}[A]$.
6. Given that, for any B, all the operators occurring in the formula $\text{Pr}[B]$ are intuitionistic, we finally obtain that $\text{Pr}[\Gamma] \models \text{I}_p \text{Pr}[A]$.

Theorem 3 (Completeness). If $\Gamma \models E_p A$, then $\Gamma \vdash NE_p A$.

Proof. Assume that $\Gamma \models E_p A$. By lemma [7], $\text{Pr}[\Gamma] \models I_p \text{Pr}[A]$. By the completeness of the system I_p , we have that $\text{Pr}[\Gamma] \vdash I_p \text{Pr}[A]$. The result now follows directly from lemma [5].

5. Related and future work

5.1 Related work

The idea of using different signs for the different meanings attached to intuitionistic and classical operators is not new, it was used by P. Krauss in 1992⁷. The same idea was used again in 2015 by Gilles Dowek⁸. Both Krauss and Dowek have classical versions of $\wedge$ and $\forall$. It is interesting to observe that (1) $\wedge_c$ does not satisfy (in general) projections and is not idempotent and that (2) $\forall_c$ does not (in general) satisfy universal instantiation. The main motivation of both Krauss and Dowek was to explore the possibility of hybrid readings of axioms of mathematical theories. The example discussed by Krauss is the axiom of choice and the example discussed by Dowek is also taken from set theory. The whole point is, in Dowek's own words, to consider that "which mathematical results have a classical formulation that can be proved from the axioms of constructive set theory or constructive type theory and which require a classical formulation of these axioms and a classical notion of entailment remains to be investigated". As an example, taken from Dowek⁹ and adapted to NE_p , we present a "hybrid" proof of the set-theoretical theorem that if the union of two sets is infinite, then at least one of them is infinite. The proof is "hybrid" in the sense that the axiom is formulated with intuitionistic operators, and the conclusion is obtained for the classical disjunction $\vee_c$.

7. Peter H. Krauss, "A constructive interpretation of classical mathematics". Mathematische Schriften Kassel, preprint No. 5/92, 1992.

8. Gilles Dowek, "On the definitions of the classical connective and quantifiers". In *Why is this a proof*, edited by Edward Hermann Haeusler, Wagner Sanz and Bruno Lopes, (UK: College Books, 2015), 228-238.

9. Dowek, *op. cit.*, 237.

$$\begin{array}{c}
\frac{\frac{\frac{\forall x \forall y (F(x) \rightarrow_i (F(y) \rightarrow_i F(x \cup y)))}{(F(a) \rightarrow_i (F(b) \rightarrow_i F(a \cup b)))} [F(a)]^1}{(F(b) \rightarrow_i F(a \cup b))} [F(b)]^2}{F(a \cup b)} \neg F(a \cup b) \\
\frac{1 \quad \frac{\perp}{\neg F(a)}}{\neg F(a)} [\neg \neg F(a)]^3 \\
\frac{2 \quad \frac{\perp}{\neg F(b)}}{\neg F(b)} [\neg \neg F(b)]^4 \\
\hline
3,4 \quad \frac{\perp}{(\neg F(a) \vee_c \neg F(b))}
\end{array}$$

5.2 Future work

1. Although we have proved the normalization for NE_p , clearly there are lots of things to be done with respect to the proof theory of the system. We know that we do not have as a corollary of normalization the sub-formula principle in its usual form, but can we have a weak sub-formula principle based on the intended meaning of the classical operators? Can we have confluence? Strong Normalization?
2. The results obtained in this paper will be extended to first order logic and to some modal logics.
3. It would be interesting to explore the intended meaning of the classical operators in order to obtain a Curry-Howard type of result for the ecumenical system.
4. As we mentioned above, an interesting application of ecumenical systems is related to the analysis of mathematical results that depend on ecumenical readings of axioms (see Krauss and Dowek¹⁰). It would certainly be interesting to pursue the investigation of other axiomatic theories.
5. We are also planning to define an ecumenical sequent calculus and an ecumenical tableaux system.

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10. See, Dowek, *op. cit.* See, Krauss, *op. cit.*